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In [1]: import numpy as np
import matplotlib.pyplot as plt

Z_values = [-2.0, -0.5, 0.0, 1.0, 3.5]

def sigmoid(z):
    return 1 / (1 + np.exp(-z))

def relu(z):
    return max(0, z)

print(f'{"Z":>6} {"Sigmoid":>10} {"ReLU":>6}')
print("-" * 28)
for z in Z_values:
    print(f"Z = {z:>4} → Sigmoid ≈ {sigmoid(z):.3f} | ReLU = {relu(z)}")

zero_relu = [z for z in Z_values if relu(z) == 0]
print(f"\nReLU outputs zero for Z values: {zero_relu}")
print("Implication: Neurons with  $Z \leq 0$  output zero gradient → 'dying ReLU' problem")
```

Z	Sigmoid	ReLU
-2.0	0.119	0
-0.5	0.378	0
0.0	0.500	0
1.0	0.731	1.0
3.5	0.971	3.5

ReLU outputs zero for Z values: [-2.0, -0.5, 0.0]

Implication: Neurons with $Z \leq 0$ output zero gradient → 'dying ReLU' problem; those neurons stop learning.

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In [2]: Z = np.linspace(-5, 5, 300)

plt.figure(figsize=(7, 4))
plt.plot(Z, sigmoid(Z), label='Sigmoid', color='steelblue')
plt.plot(Z, np.maximum(0, Z), label='ReLU', color='tomato')
plt.axhline(0, color='black', linewidth=0.6, linestyle='--')
plt.axvline(0, color='black', linewidth=0.6, linestyle='--')
plt.title('ReLU vs Sigmoid Activation Functions')
plt.xlabel('Z (Input)')
plt.ylabel('Activation Output')
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```

ReLU vs Sigmoid Activation Functions

